

Communication Access Real-Time Translation Through Collaborative Correction of Automatic Speech Recognition

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Abstract

Communication access real-time translation (CART) is an essential accessibility service for d/Deaf and hard of hearing (DHH) individuals, but the cost and scarcity of trained personnel limit its availability. While Automatic Speech Recognition (ASR) offers a cheap and scalable alternative, transcription errors can lead to serious accessibility issues. Real-time correction of ASR by non-professionals presents an under-explored CART workflow that addresses these limitations. We conducted a user study with 75 participants to evaluate the feasibility and efficiency of this workflow. Complementary, we held focus groups with 25 DHH individuals to identify acceptable accuracy levels and factors affecting the accessibility of real-time captioning. Results suggest that collaborative editing can improve transcription accuracy to the extent that DHH users rate it positively regarding understandability. Focus groups also showed that human effort to improve captioning is highly valued, supporting a semi-automated approach as an alternative to stand-alone ASR and traditional CART services.

CCS Concepts

• **Human-centered computing** → **Accessibility technologies**; *Empirical studies in accessibility*; *Accessibility design and evaluation methods*; • **Computing methodologies** → **Speech recognition**.

Keywords

live captioning; subtitles; deaf and hard of hearing; accessibility;

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1 Introduction

Transcription plays a key role in providing d/Deaf and hard of hearing (DHH) individuals with access to spoken language. Real-time captioning presents significant challenges as text must be produced almost synchronously with speech. Professional services, known

as Communication access real-time translation (CART), typically rely on stenography or re-speaking (voice-writing). However, the scarcity and high cost of trained transcribers limit the availability of such services and, thus, the accessibility for many events. Advances in Automatic Speech Recognition (ASR) have greatly improved its accuracy, offering a scalable and cost-effective alternative [10, 12, 19, 43, 45]. As a result, automated captioning is now integrated into widely used applications and is increasingly relied upon as an accessibility service [42].

At the same time, DHH individuals have consistently reported serious accessibility issues for automatically generated captions [8, 13, 25, 48] and have criticised the lack of a declarative ruling regarding the quality of ASR [9, 48]. There seems to be a disparity between measured ASR accuracy and DHH individuals' experience. One reason for this discrepancy is that benchmark scores can be misleading [16]. The performance of an ASR model in practice heavily depends on how well its training data cover the actual conditions [27]. While high accuracy is achieved for General American English, noisy recording conditions [26], specialised vocabulary [45], or a speaker's accent [54] can significantly decrease the performance, particularly in lower-resourced languages [5]. Furthermore, commonly used evaluation metrics offer a limited representation of accessibility, leading to a mismatch between measured accuracy and the perceived transcription quality [14, 38, 59]. Until ASR systems reliably produce accessible captions, CART services involving humans are essential, and alternatives that address their limitations must be explored.

One promising yet under-explored approach to improving transcription accuracy is the manual correction of ASR output [28, 37, 52, 57]. As ASR becomes more robust to noise and speaker variability, re-speaking benefits may diminish, allowing users to focus on corrections. This approach reduces the cognitive load on transcribers and could be performed by less trained personnel. This paper uses a mixed methods approach to evaluate a CART workflow in which a group of non-professionals correct ASR output in real-time to increase transcription accessibility. We developed a prototype that combines a live video stream, ASR output, and collaborative text editing. A user study evaluates the feasibility and transcription accuracy of the approach. Complementary, DHH focus groups were conducted to correlate their accuracy ratings with the user study results and to align their preferences with the capabilities of the workflow in general. Our results show:

- Corrections can improve ASR accuracy to the extent that DHH readers rate it positively regarding understandability.
- Common evaluation metrics ignore quality aspects that focus groups consider equally important to word accuracy.
- The editing task is demanding and limits knowledge transfer.

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2 Background

2.1 Automatic Speech Recognition

ASR accuracy has improved significantly in recent years, with state-of-the-art (SOTA) models achieving high accuracy in many languages [5, 45]. This progress has largely been driven by adopting end-to-end (E2E) models, particularly transformer architectures [10, 12, 19, 43, 56], and machine learning techniques. Unsupervised [62], semi-supervised [47], and self-supervised [51] learning approaches use web-scale training data to improve the robustness and overall quality of ASR systems. Regarding accuracy rates, it has been suggested that ASR has reached parity with manual transcription [61]. However, these results can be misleading, as optimising for specific datasets comes with the risk of shortcut learning [16]. Despite improvements, the performance of ASR can be limited by poor audio quality [26], low-resource languages [5, 45] and biased training data [27, 54].

Studies indicate that the accuracy of automated captioning is insufficient to provide DHH readers with full access to spoken content [8, 13, 25]. The National Association of the Deaf (NAD) has reported significant issues with captioning accuracy, timing, and completeness, and is seeking a declaratory ruling to address these concerns [48]. Furthermore, the readability of ASR-generated captions can be challenging [8, 34], especially when captions are not syntactically segmented [17, 29]. Streaming ASR further complicates the experience by producing intermediate results that cause distracting text changes in live captions [35].

2.2 Manual Correction of ASR

Correcting ASR effectively reduces error rates [11] and has also been suggested by DHH individuals [37, 52]. This method is already used by re-speakers, who use ASR and manually correct the remaining errors. Correction is more efficient than creating a full manual transcription when the error rate is below 30% [15], a threshold that ASR models can reliably achieve [1, 45]. While errors can be reduced by 30% to 50% [11, 40, 53], transcription quality can be higher as users tend to correct errors that have a greater impact on comprehension [7, 30]. DHH readers also rated manually corrected captions highly accurate, sometimes preferring them over professional transcription services [13].

Real-time transcription is particularly challenging, as speech is up to ten times faster than typing [58], and corrections must be made quickly [46]. When working with non-professionals to overcome the shortage of professional transcribers [32], distributing the task among multiple individuals can reduce the demand and increase quality [20]. Studies indicate that at least three non-expert typists are required to correct ASR transcriptions [55], reducing errors by around 30% [28]. While real-time correction of ASR has been proposed to increase accessibility [37, 52, 57], further research is needed to assess its feasibility and effectiveness [13, 28, 30].

2.3 Measuring Transcription Quality

The accuracy of ASR transcriptions is commonly measured by the Word Error Rate (WER), which uses the Levenshtein distance on a word basis to determine the minimum edits between two texts [33]. The WER is sensitive to non-semantic differences, treating them

equally to missing or incorrect words. To address this, texts are often normalised by removing punctuation and converting to lowercase. Further steps, like unifying abbreviations, contractions, and written numbers, can reduce non-semantic errors [27, 45]. However, these modifications may remove aspects that affect comprehension, such as punctuation or semantic formatting.

The WER has been criticised as a measure of accessibility, as it does not necessarily reflect human perception of accuracy [14, 38, 59]. Alternative metrics have been proposed but have not been widely adopted yet [3, 24, 39, 50]. A key problem is that WER does not consider the contribution of individual words to the overall understandability of a text [6]. Machine learning (ML)-based metrics address this issue by classifying the importance of words to determine the severity of errors [2, 22]. While some studies report a higher correlation of such metrics with human perceptions of quality [23, 24], adoption to other domains may be limited [60], and metrics may not necessarily perform better than the WER [4]. Alternatives to automated metrics rely on manual classification of transcription errors [3, 49]. The usefulness of a transcript can also be assessed through subjective ratings, e.g. by ranking the understandability of a text. While qualitative measures are helpful for specific evaluations, they are difficult to obtain for large datasets and lack objectivity and replicability.

In this study, we use the WER as it is the standard metric in speech recognition research and to report accuracy with an objective measure. As it has received criticism as an accessibility metric, we additionally include accuracy ratings from a diverse group of DHH individuals.

3 Methodology

We used a mixed methods approach to evaluate the proposed CART workflow: Firstly, we conducted a user study in which non-professionals corrected ASR-output in real-time. We measured the accuracy and tested different collaboration scenarios to gather information on efficacy, usability, and user experience. Secondly, we conducted focus groups with DHH individuals to align their accuracy ratings with the results of the user study and their captioning requirements with the capabilities of the ASR correction workflow.

3.1 Recording

The recording is an animated video which presents a scientific topic entertainingly.¹ We assume all participants have some knowledge about the general topic, but it is unlikely they are familiar with the specific content of the video. The recording lasts 10:14 minutes, and the reference transcript contains 1381 words, an average of 135 words per minute (wpm). We used a German video, as all participants were native speakers. The audio is a high-quality recording of a professional male speaker, with soft background music and sound effects. The speaker is not visible, and no lip reading is possible. The speech is grammatically correct, with clear pronunciation.

The video was transcribed with ten different ASR models (see Appendix A). To increase the robustness of the WER, we used a custom normaliser for German (see Appendix B), inspired by a related study [45]. To start from a reasonable level of accuracy for the editing task, we selected an average transcript with a WER of

¹<https://www.youtube.com/watch?v=awwCj4NnGdk>

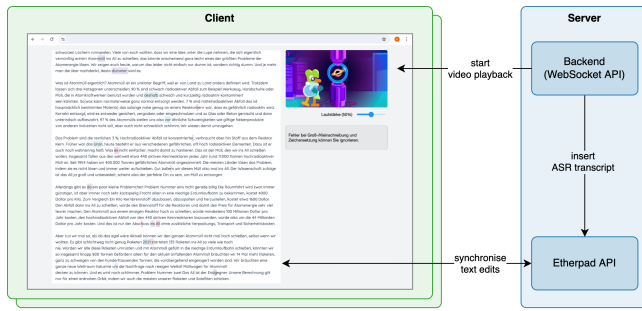


Figure 1: Architecture and user interface of the prototype

9.3%. The transcript was automatically formatted (see Appendix C). The video was split into four evenly long parts for the DHH groups’ caption rating. The parts contained captions with a WER of 2%, 5%, 10%, and 15% (see Appendix A). The transcripts were then automatically formatted into captions (see Appendix D).

3.2 User Study

3.2.1 Participants. Overall, 75 people participated in the user study. Most were undergraduate computer science students, and the other participants were postgraduates and researchers at the same university. They gave informed consent to participate in the study and received a small thank-you gift. Participants were native German speakers and did not have a hearing disability. Participants worked in groups of three to edit the captions. 60 participants were randomly assigned to the different collaboration scenarios and editing roles by signing up for a specific time slot. An additional 15 participants were assigned to a control group.

3.2.2 Collaboration Scenario. Based on the results of previous research [28, 44], we chose four collaboration scenarios to evaluate the impact of temporal and spatial task distribution on correction efficiency and editing conflicts. Temporal task distribution offsets each user’s position in the transcript by delaying audio playback. Although latency must be kept to a minimum in real-time transcription, we wanted to evaluate the extent to which delayed corrections could improve accuracy. In some streaming scenarios, for example where users don’t interact with the presenters, near-live may be tolerable [17]. Spatial task distribution mainly reduces user conflicts by assigning them different text segments. We set a group size of three, as suggested by related research [55]. For each scenario, we had five groups (i.e. 15 participants):

- A Parallel: No task distribution mechanism has been applied.
- B Delayed: Each user received the audio with a different delay (0, 10, 20 seconds).
- C Chunked: Users were assigned chunks of the transcript and took turns correcting them.
- D Mixed: Two users took turns correcting while the third user proofread with a 10-second audio delay.

The scenarios were designed to help users distribute the work, but they did not prevent users from breaking out of their intended text sections. Editors could still help to correct other users’ sections.

3.2.3 Prototype. The interface has an interactive text document on the left and a video player on the right (see Fig. 1). In contrast to caption-based editing, our interface enables continuous text editing, allowing users to collaborate more freely [28]. The design is functional and follows common caption editors and text editing solutions. An Etherpad is used to enable collaborative editing with real-time updates. The video player has only volume controls, and the server controls playback. The ASR transcript is inserted consecutively and word by word, in sync with the audio track. Text insertions by the server are shown in the default black font colour. Text changes by users are highlighted in a user-specific colour to visually indicate the other participants’ edits.

3.2.4 Procedure. Participants scheduled a time for the user study in a computer lab or remotely via video conference. They used provided hardware or their own devices. Participants were informed about the study’s goals, provided consent, and received a unique user ID for the editing software and questionnaire. All participants accessed the web application and completed a technical check to verify functionality. They were briefed on the collaboration scenario and instructed to focus on correcting word errors, ignoring capitalisation and punctuation. The video played at normal speed without interruption, and users edited the transcript without communicating with each other. After the video, participants completed a questionnaire including the NASA Raw Task Load Index (RTLX) for cognitive load [18, 21], questions on knowledge transfer from the video, and usability and user experience evaluations. They were then debriefed and given a small thank-you gift.

3.2.5 Control Group. To classify editing users’ cognitive load and knowledge transfer, a control group of 15 hearing participants watched the video without captions. After watching the video (without the option to rewind it), they answered a questionnaire with the RTLX and the same questions regarding the videos’ content as the editing participants.

3.3 DHH Focus Groups

25 DHH individuals using assistive technologies like hearing aids or cochlear implants participated in three focus groups. Two groups were local DHH interest groups, one using spoken language and the other sign language, with diverse ages and occupations. The third group included university students from a DHH initiative. Participants provided informed consent and received financial compensation. Despite diverse impairments, participants could be divided into two groups: those who relied solely on captioning and those who used captioning in addition to hearing. Focus groups were held in the groups’ usual meeting places, with sign language interpreters assisting the sign language group.

Sessions began with a modified experience interview [63], where participants shared positive live captioning experiences using a handout for guidance. These experiences led to group discussions, which were transcribed and analysed for recurring themes and experiences [36]. Participants watched the same video used for the editing’ study in four parts, each with captions at different accuracy levels. The video was shown on a projector with sound available in the room and via an induction loop. After each part, participants rated caption understandability on a seven-point Likert scale and

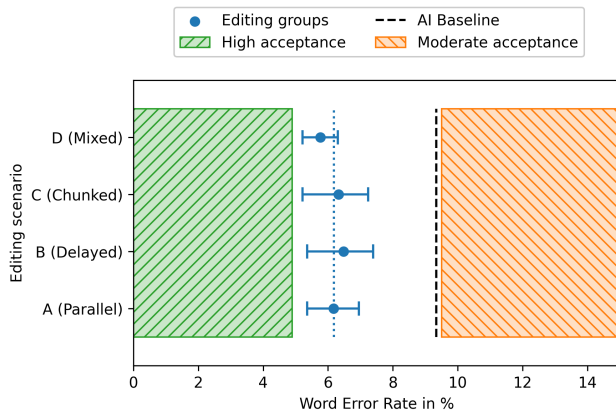


Figure 2: Word Error Rate by editing scenario

gave written feedback. Final ratings and comments addressed accuracy, punctuation, formatting, latency, and verbatim transcription. Sessions concluded with debriefing.

4 Results

4.1 Transcription Accuracy

Fig. 2 shows the WER of the ASR transcript, the WER after the corrections of each group, and the WER ranges of the focus groups' acceptance ratings. The initial WER was 9.3%. On average, the groups' modifications resulted in a WER of 6.2% ($SD = 0.7\%$, $min = 5.2\%$, $max = 7.4\%$). A Kruskal-Wallis test was conducted to determine whether there is an effect of the editing scenario on the WER after correction. There was no significant difference in the resulting WER between the four different editing scenarios [$H(3) = 2.759$, $p = .430$] with a mean WER score of 6.2% for scenario A, 6.5% for scenario B, 6.3% for scenario C and 5.8% for scenario D. On average, punctuation errors were reduced by 23% ($SD = 15\%$), and capitalisation errors by 15% ($SD = 16\%$).

4.2 Editing Users' Comments

Editing users' comments on the questionnaire were categorised as positive or negative, and grouped into three topics: related to the general task, the usability of the prototype, and collaboration within the group. In total, 115 comments were negative, and 101 were positive. The number of comments grouped by topic was: task (negative = 56, positive = 28), usability (negative = 31, positive = 13), collaboration (negative = 28, positive = 60). 31 participants perceived the task as stressful or demanding, while 13 mentioned terms like "quite relaxing" or "rather easy". 11 participants perceived correcting mistakes as rewarding. 37 participants commented positively on teamwork, 6 negatively, and 5 perceived teamwork as competitive. 16 participants mentioned editing conflicts with other team members, and 15 in following the continued video after correction.

4.3 Cognitive Load

The average RTLX value for the editing users was 44.4 ($SD = 17.6$, $Mdn = 45.4$), and for the control group users was 13.4 ($SD = 11.2$, $Mdn = 12.5$). A Kruskal-Wallis test was conducted to determine

whether there is an effect of the editing task on the perceived cognitive load. The results show a significant difference [$H(4) = 29.693$, $p < .001$]. A pairwise post-hoc Dunn-Bonferroni's test found significant differences between participants in scenarios A, B, and C ($p < .001$) and D ($p = .023$), indicating that the editing groups perceived the task as significantly more stressful than the control group. No other differences were statistically significant.

4.4 Knowledge Transfer

On average, editing users answered 4 questions correctly ($SD = 1.5$, $Mdn = 4.0$) and control groups answered 5.5 questions correctly ($SD = 1.1$, $Mdn = 6.0$). A Mann-Whitney U test was conducted to determine whether there is a difference in the number of correct answers for editing participants and control group participants. There was a significant difference between the participant groups ($U = 202.5$, $p < .001$). Editing users rated their ability to follow the content on a seven-point Likert scale from 1 (very bad) to 7 (very good) on average with 3.6 ($SD = 1.5$, $Mdn = 3.0$). We analysed participants' answers to how well they could follow the content while editing. 33 participants reported they could not follow the content at all, 14 participants could partly follow, and 9 could follow. 22 participants commented that they focused on hearing and reading to detect errors, not the actual content. 25 participants reported it was difficult or impossible to follow the content while correcting errors (or for a short time after a correction).

4.5 Focus Groups' Ratings

Participants rated the understandability of four video chunks with different WER levels on a seven-point Likert scale from 1 (not understandable) to 7 (perfectly understandable). One participant did not fill out all scales and was excluded. A one-way ANOVA was conducted to determine the effect of the WER on participants' subjective rating of understandability. There was a significant effect of the video's WER on participants' rating, indicating that participants rated a lower WER more positively for understandability [$F(3, 92) = 8.710$, $p < .001$]. A Tukey's HSD test revealed a significant difference between participants' rating for video between a WER of 4.9% and 14.7% ($p = .006$), between a WER of 2.2% and 14.7% ($p < .001$), between a WER of 4.9% and 9.5% ($p = .014$), and between a WER of 2.2% and 9.5% ($p = .002$).

Participants rated the importance of different quality aspects of captions on a seven-point Likert scale from 1 (not important) to 7 (very important): accuracy ($M = 6.2$, $SD = 1.2$, $Mdn = 7.0$), punctuation ($M = 5.5$, $SD = 1.4$, $Mdn = 6.0$), capitalisation ($M = 5.9$, $SD = 1.6$, $Mdn = 7.0$), formatting ($M = 5.2$, $SD = 1.8$, $Mdn = 6.0$), latency ($M = 5.4$, $SD = 2.1$, $Mdn = 7.0$) and how verbatim captions have to be ($M = 5.6$, $SD = 2.2$, $Mdn = 7.0$).

5 Discussion

We tested an alternative CART workflow in which non-professionals collaboratively correct ASR. We also conducted focus groups with DHH individuals to compare their preferences for live captioning with the semi-automated transcription process. We measured caption quality using the WER and with ratings from DHH caption readers, as there is currently no objective measure for the accessibility of captions.

All editing groups could reduce the WER of the ASR transcript. The initial WER was 9.3%, a level that reached medium acceptance by the DHH focus groups. The corrected transcripts achieved an average WER of 6.2%, similar to the accuracy range rated positively regarding understandability. The relative error reduction of 33% is consistent with findings from related studies [11, 28, 40, 53]. Although users were instructed to focus on word errors, punctuation and capitalisation errors also decreased, suggesting that people tend to correct more meaningful errors, which are not measured by WER but enhance comprehension (cf. [7, 30]). The comprehensibility of the transcript may therefore be higher than the WER suggests.

The collaboration scenario had no significant effect on the resulting accuracy of the transcript or the perceived cognitive load of the users. While teamwork was perceived as stress-reducing, a mechanism to guide collaboration seems necessary, as most users within the parallel scenario reported editing conflicts. Editors perceived working in chunks as more positive, as it reduced such conflicts and allowed them more time to review their segment of the text. Since working with delays did not improve performance, and the DHH focus groups emphasised their preference for minimal latency (cf. [37]), temporal task distribution seems to be neither an effective nor a viable solution. Correcting mistakes was perceived as rewarding, but participants also mentioned a sense of competition. A revised interface could emphasise the task's collective goal, for example, by displaying the total number of corrections made.

The editor's average RTLX score was 44, which is not overly demanding but does require some effort [18]. The score was significantly higher than that of the control group, who only watched the video. In the quiz following the video, editors gave significantly fewer correct answers than participants in the control group. Most editors reported that they could not or only partially follow the content, especially whilst or shortly after correcting errors. This suggests that following the content of the video and correcting errors in the script is too demanding to be done simultaneously [31]. Therefore, the task of editing should be performed by dedicated editors so as not to disadvantage other listeners [37].

Reading error-free captions is already a demanding task because the speed of speech often exceeds that of reading. Any mistakes, such as word errors, missing punctuation, incorrect capitalisation, or poor formatting, disrupt the reading flow and require mental compensation that can only be managed to a limited extent [41]. Acceptance ratings among DHH users varied significantly depending on the WER; captions with 2-5% WER were rated significantly higher than those with 10-15% WER (cf. [8]). Positive comments were more frequent for captions with fewer errors, while higher error rates led to more negative feedback. The impact of errors on reading speed is also reflected in the DHH participants' negative comments on talking speed in video segments with higher WER, despite similar wpm rates across segments.

DHH individuals rated accuracy, capitalisation, punctuation, and semantic formatting as equally important for understandability (cf. [8, 29]), supporting the criticism of WER as a limited measure of accessibility [4, 14, 38, 59]. The importance of different quality aspects is strengthened by the fact that captions with lower WER also had fewer punctuation and capitalisation errors. Minimal latency was strongly preferred, as caption reading is often combined with lip-reading or hearing (cf. [37]). Focus groups also expressed a clear

preference for verbatim transcription. While ASR systems can provide low-latency verbatim output, corrections remain essential to address errors that hinder comprehension. Combining the speed of ASR with human editing is a promising alternative to traditional CART workflows to improve accessibility until fully automated systems can reliably meet user needs.

Participants expressed predominantly negative experiences with ASR-generated captions and their preference for professional caption services: *"I can clearly see whether it is people who make an effort to write everything down or whether only ASR is used"*. ASR was seen as frustrating and unreliable, described as *"very irritating"* and producing *"wondrous linguistic creations"*. In contrast, human involvement was highly valued. Participants stated, *"A human being should still stand between AI and us"*, and found their transcription errors more tolerable: *"Mistakes made by written interpreters are forgivable"*. The discussions showed that automated captioning lacks accessibility, while human effort is highly valued. An ASR correction workflow could bridge this gap and provide a more acceptable solution for DHH readers.

6 Conclusion

ASR offers a great opportunity to improve the accessibility of events through real-time captioning, but its lack of accuracy limits the participation of people who rely on it. Collaboratively correcting ASR could improve accessibility, especially when professional transcribers are unavailable. We evaluated this alternative CART workflow in a user study and conducted focus groups to align its capabilities with the preferences of DHH individuals.

The user study showed that the initial WER of ASR can be reduced to the range of high acceptability, even though corrections must be made in real-time. As humans tend to correct errors that influence the meaning of a text, the accessibility may be even higher than the WER indicates. As real-time correction of ASR is demanding, it requires dedicated personnel apart from the audience to ensure that knowledge transfer is not hindered. Correcting as a group can make the task more manageable, and spatial task distribution can reduce stress and editing conflicts. While relying solely on ASR may feel like a second-rate accommodation for DHH individuals, human efforts to improve their participation were highly valued, potentially increasing the acceptance of ASR in a semi-automated captioning workflow.

7 Limitations & Future Work

Our findings are based on a single video presentation with a professional speaker, limiting generalisability to other contexts. Less favourable conditions impact ASR accuracy, complicate manual correction, and reduce intelligibility for caption readers. DHH participants did not rate the edited captions directly to avoid bias from individual editing groups. At this stage of the research, we believe that comparing the WER after editing with the WER ratings of DHH individuals is a more objective and transferable approach to other studies. Future research should evaluate the workflow end-to-end, with editors correcting streaming ASR and DHH users reading and rating the corrected transcript. It could also be explored how AI models can assist humans by providing autocompletions, suggestions or autonomous corrections.

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- ## A ASR Transcriptions

Table 1 displays the WER with text normalisation for each video chunk and the whole transcript. Bold marked entries were used for the user study (overall) and the DHH readers' caption rating (per chunk).

B Text Standardisation

The transcripts were automatically normalised before calculating the WER according to the following rules:

Table 1: Word Error Rate in percent by ASR model

Service	Overall	Chunk 1	Chunk 2	Chunk 3	Chunk 4
Amazon	8.62	7.06	9.70	9.46	8.10
AssemblyAI	8.91	10.59	11.59	7.45	5.61
Deepgram	16.00	15.88	17.25	13.18	17.76
Google	12.96	14.71	12.13	10.32	14.95
IBM	16.00	17.35	15.63	11.17	20.25
Microsoft	6.37	7.06	8.63	5.44	4.05
Rev AI	9.34	9.12	9.97	9.46	8.72
Speechmatics	3.69	2.65	4.85	4.30	2.18
Speechtext.AI	11.08	10.88	15.36	8.60	9.03
Whisper	4.06	7.94	2.70	3.44	2.18
Average	9.70	10.32	10.78	8.28	9.28

- (1) Replace all white space characters with a space and remove repeated spaces.
- (2) Replace German diacritics (e.g. ä -> ae, ß -> ss).
- (3) Convert currencies and symbols to their written forms (e.g. € -> Euro, % -> Prozent).
- (4) Replace common German abbreviations (e.g. z.B. -> zum Beispiel).
- (5) Convert common German contractions to its long form (e.g. gibt's -> gibt es).

- (6) Convert all Arabic numbers to a numeric expression (e.g. 200 -> zweihundert).
- (7) Transform text to lowercase and remove all punctuation.
- (8) Remove repeated, leading, and trailing spaces.

C Transcript Formatting

The transcript was automatically formatted according to the following rules:

- Start a new paragraph after a silence of 5 seconds.
- 800 to 1000 characters per paragraph (100 to 200 for chunked scenarios).
- End paragraph on sentence ending characters (.!?:).

D Caption Formatting

The captions were automatically formatted according to the following rules:

- Length of 60 to 80 characters.
- End caption if a sentence ending character appears (.!?:).
- Format the caption into two lines:
 - Break in between the character ranges of 30 to 60.
 - Try to break at a punctuation character (.,!?:).
 - Otherwise, break at any non-word character (e.g. white space).